

Intergenerational Income Mobility and Private Education in South Korea: A Varying-Coefficient IGE Analysis with Regional Heterogeneity

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Abstract

Step 1 established three robust empirical patterns for South Korean intergenerational income mobility using KLIPS 1998–2024 ($N = 5,388$): the Capital Region’s intergenerational earnings elasticity (IGE) substantially exceeds that of metropolitan and provincial regions; bottom-quantile IGE is nearly three times the top-quantile IGE; and daughters’ IGE is approximately twice sons’ IGE. Step 2 asks why these regional gradients exist and how they are sustained by regional private education markets and university entrance examination (CSAT/) competition. The core estimating framework is a varying-coefficient IGE regression: subtracting initial log income from both sides of the standard IGE equation yields a growth-rate form that is algebraically equivalent to the cross-regional regression of Durlauf, Kourtellos, and Minkin (2001). We adopt DKM’s varying-coefficient estimation strategy — not its structural growth-model interpretation — with regional average CSAT scores as the smoothing variable rather than initial per-capita income, to test whether the intergenerational earnings elasticity varies systematically across regions with different levels of exam competition. Inference relies on wild cluster bootstrap (Cameron, Gelbach, and Miller, 2008) throughout, given the small number of regional clusters ($J = 12$ – 17). An individual-level pooled OLS with region-clustered standard errors and a quantile regression extension by regional group complete the analysis. Hierarchical linear models are not used, owing to concerns about distributional assumptions on random effects with only $N = 17$ regional units; all cross-level interactions are instead estimated via pooled OLS interaction terms with clustered standard errors.

Motivation

Step 1 of this thesis documented three empirical patterns from the Korean Labor and Income Panel Study (KLIPS, waves 1–27, 1998–2024) that serve as the point of departure for the analysis that follows. First, the Capital Region’s intergenerational earnings elasticity (IGE) substantially exceeds that of metropolitan and provincial regions: Seoul registers an IGE of 0.178, metropolitan cities average approximately 0.13, and provincial regions approximately 0.08. Second, intergenerational persistence is highly asymmetric across the income distribution: the bottom-decile IGE (0.161) is nearly three times the top-decile IGE (0.057). Third, daughters exhibit approximately twice the IGE of sons (0.181 versus 0.095), a finding absent from the prior Korean literature, which focused almost exclusively on sons. Taken together, these patterns suggest that the intergenerational transmission of economic status in Korea is not a uniform national phenomenon but instead varies systematically by region, by income level, and by gender.

The structural story motivating Step 2 is the following. Parental income determines the intensity of private tutoring investment, which in turn determines performance on the university entrance examination (CSAT,), which determines placement at a selective university, which ultimately determines the child’s adult earnings trajectory. Because private tutoring academies (*hagwons*) are heavily concentrated in the Capital Region — Seoul and Gyeonggi account for a disproportionate share of national tutoring expenditure — the regional concentration of this mechanism provides a natural explanation for the spatial IGE gradient documented in Step 1. Step 2 tests this mechanism rigorously using a varying-coefficient IGE regression, borrowing the estimation strategy of Durlauf, Kourtellos, and Minkin (2001), which allows the intergenerational earnings elasticity to differ across regions with different levels of exam competition. The central research question is: *why does intergenerational income mobility differ across Korean regions, and how do regional CSAT performance and private education intensity mediate this variation?*

Literature Review

Intergenerational Income Mobility: Theory and Measurement

The canonical theoretical framework for intergenerational income mobility is the human capital investment model of Becker and Tomes (1979, 1986). In their formulation, high-income parents invest more in children’s education, health, and social networks, generating a positive correlation between parent and child permanent income. The magnitude of this correlation — the IGE — reflects both the productivity of parental investment and the degree to which capital market imperfections constrain lower-income families from investing optimally. This framework leads to the prediction that where credit constraints are most binding, at the lower end of the income distribution, intergenerational persistence should be strongest — a prediction Step 1 confirmed through quantile regression.

Galor and Zeira (1993) extend the Becker–Tomes framework by introducing a non-convex production technology that generates a poverty trap: below an income threshold, families cannot borrow to finance optimal human capital investment regardless of expected returns, while above the threshold investment is unconstrained. Their model predicts, first, that the IGE is strictly higher for credit-constrained families (consistent with the Step 1 finding that bottom-quantile IGE is nearly three times the top-quantile IGE), and second, that regions where private education markets are financially inaccessible to lower-income families will exhibit higher aggregate IGE. Both predictions directly motivate the Step 2 design. In Korea’s institutional context, private tutoring markets represent the primary channel through which parental income is converted into human capital investment, and the geographic concentration of those markets makes the Galor–Zeira mechanism spatially heterogeneous in exactly the way Step 2 seeks to quantify.

The standard econometric framework for IGE estimation is the log-log regression (Solon, 1999), $\ln y_i^c = \alpha + \beta \ln y_i^p + \varepsilon_i$, where $\hat{\beta}$ is the IGE. Haider and Solon (2006) show that observing income at non-optimal ages introduces lifecycle bias whose severity varies with the age of observation and, crucially, with the occupation and income level of the parent, since lifetime earnings profiles peak at different ages for different occupational groups. In the Korean context this is directly relevant: Seoul high-income professional parents tend to peak around ages 45–55, while provincial manufacturing workers peak earlier at roughly 40–45. This means Seoul parental income is relatively more undermeasured in the KLIPS panel, implying that the estimated “Capital > Province” IGE gradient is a conservative lower bound on the true gradient. Solon (2004) develops a general model decomposing the IGE into the heritability of productive traits, the return to education, and the progressivity of public investment in children, a decomposition that motivates the separation of public and private education spending in the growth-rate regression below.

From IGE to Growth-Rate Form: The Statistical Foundation

The theoretical starting point is the generalized IGE regression, in which the intergenerational earnings elasticity is allowed to vary with a region-level characteristic Z_j (regional CSAT competi-

tion intensity):

$$\ln y_{i,t} = b(Z_j) \ln y_{i,t-1} + \varepsilon_{i,t}.$$

Subtracting $\ln y_{i,t-1}$ from both sides and defining the intergenerational income growth rate $g_{i,t} \equiv \ln y_{i,t} - \ln y_{i,t-1}$ yields:

$$g_{i,t} = [b(Z_j) - 1] \ln y_{i,t-1} + \varepsilon_{i,t} \equiv c(Z_j) \ln y_{i,t-1} + \varepsilon_{i,t},$$

where $c(Z_j) \equiv b(Z_j) - 1 < 0$ whenever $b(Z_j) < 1$ — the standard mean-reversion condition. This growth-rate form is algebraically identical to the cross-regional regression of Durlauf, Kourtellos, and Minkin (2001), $g_j = \gamma(z_j)' \mathcal{X}_j + \varepsilon_j$, with the coefficient on initial income playing the role of the varying convergence parameter. The DKM framework is therefore adopted for its **estimation strategy** — a varying-coefficient regression on a growth-rate equation — not for the structural content of the Solow model.

In the regional implementation, additional controls enter the growth-rate regression to account for compositional differences across *sido*. Public and private human capital expenditures are kept strictly separate throughout, because they have opposite distributional properties: public education spending per pupil in Korea is effectively *regressive* — government equalization grants allocate more funds to provincial regions with smaller enrollment — while private tutoring expenditure is strongly *progressive*, co-moving with household income. Combining the two into a single human capital variable creates sign cancellation, producing a near-zero composite coefficient not because education is irrelevant but because the regressive and progressive components offset each other. The correct specification keeps h_j^{pub} and h_j^{priv} as separate regressors throughout.

The diagnostic value of the growth-rate regression lies in its residual plot. If the IGE coefficient $b(Z_j)$ is genuinely constant across regions (the fixed-coefficient null), the residuals from a constant-coefficient growth regression will be unpatterned against initial income. A systematic positive residual pattern for high-CSAT, high-income regions and negative residuals for low-CSAT regions constitutes visual evidence that $b(Z_j)$ — and therefore $c(Z_j)$ — varies with regional exam competition, motivating the varying-coefficient specification.

The Local Solow and Varying-Coefficient Models

Durlauf, Kourtellos, and Minkin (2001) propose the Local Solow model, in which all growth parameters are allowed to vary nonparametrically with a smoothing variable z_j : $g_j = \gamma(z_j)' \mathcal{X}_j + \varepsilon_j$, where $\mathcal{X}_j$ is the vector of MRW regressors and $\gamma(z_j)$ is an unknown smooth function of z_j . Their key finding is that growth clubs — groups of countries sharing similar structural parameters — emerge naturally from the data when z_j is initial per-capita income, without imposing an *a priori* classification. This paper applies the DKM framework to Korean regional data, using it first as a diagnostic for parameter heterogeneity and then as the estimation strategy for the varying-coefficient IGE regression.

There is a critical methodological subtlety in choosing the smoothing variable that must be ad-

dressed explicitly. In DKM’s original application, $z_j = \ln y_{j,0}$, which is also one of the explanatory variables in $\mathcal{X}_j$. Under the parametric first-order approximation $\beta_k(z_j) \approx \beta_k + \gamma_k z_j$, the “varying convergence coefficient” term becomes $(\beta_1 + \gamma_1 z_j) \ln y_{j,0} = \beta_1 \ln y_{j,0} + \gamma_1 (\ln y_{j,0})^2$, which is simply a quadratic term in initial income. This is not genuine parameter heterogeneity — it is a polynomial expansion of the baseline regression. Durlauf himself would identify this immediately: if the smoothing variable and an explanatory variable are identical, the varying-coefficient structure and a polynomial regression are mathematically equivalent, and no test can distinguish them. The solution adopted in this paper is to use $z_j = \overline{\text{CSAT}}_j$ (regional average CSAT score) as the smoothing variable for the main specification. This ensures the smoothing variable is distinct from all regressors and is theoretically motivated: the IGE coefficients should vary with the level of exam competition, since CSAT performance determines access to the selective universities that deliver the largest earnings premia, and this mechanism should operate more powerfully in regions where exam competition is more intense. The DKM original specification with $z_j = \ln y_{j,0}$ is retained as a robustness check. If both specifications yield qualitatively consistent results, the polynomial equivalence concern does not overturn the finding.

Shadow Education, CSAT, and Intergenerational Mobility

Korea’s private tutoring sector — colloquially *sagyoyuk* () — represents the world’s most intensive use of formal out-of-school instruction as a preparation vehicle for competitive university admission. Participation rates exceed 70 percent of all primary and secondary school students nationally (Statistics Korea, 2022), and expenditure is highly concentrated among upper-income households. The CSAT is a single nationally administered examination taken each November by virtually all high school seniors, and its score is the primary determinant of university placement. Because a handful of highly selective universities — informally grouped as *SKY* (Seoul National, Yonsei, Korea) plus a small number of other top-tier institutions — offer earnings premia substantially above those of other four-year institutions, the CSAT functions as the central bottleneck through which parental income, transmitted via private tutoring investment, is converted into children’s earnings prospects.

Lee (2021) provides evidence of a glass floor effect in Korean higher education: high-income families use credentialing strategies — of which tutoring is the primary instrument — to protect their children’s relative position even when the absolute return to education declines. Ku, Lee, and Kim (2022) show that the relationship between tutoring and educational inequality is nonlinear: the marginal effect of an additional unit of tutoring expenditure on test performance is highest for students near the margin between university tiers, a prediction consistent with the varying-coefficient structure in which the private education effect on IGE is strongest in regions where the marginal CSAT point most sharply determines university placement. Choi and Park (2016) document that regional variation in private tutoring intensity in Korea corresponds closely to regional variation in educational stratification, providing direct motivation for the cross-regional design.

The key distinction between this paper and the prior shadow education literature is the unit of analysis and the outcome variable. Prior studies examine individual tutoring expenditure as the treatment and individual test scores or university enrollment as the outcome. This paper uses regional tutoring market intensity as a moderator of intergenerational income persistence, testing whether the structural relationship between parents’ and children’s economic status is amplified by the presence of a concentrated private tutoring market in the region. This is a fundamentally different and arguably more policy-relevant question, because it captures the equilibrium effects of tutoring market concentration on social mobility that no individual-level study can identify: even if a family does not itself use tutoring, living in a region where most families do may raise the competitive threshold for university admission and thereby amplify the disadvantage of lower-income families.

Inference with Small Clusters

A recurring methodological challenge in regional economic analyses is that the number of regional units — the clusters — is small. With $J = 17$ *sido* (Korean metropolitan cities and provinces), standard cluster-robust standard errors are known to be downward-biased, leading to systematic over-rejection of null hypotheses (Cameron and Miller, 2015). Cameron, Gelbach, and Miller (2008) propose the wild cluster bootstrap as a solution that delivers correct size even with as few as 10–20 clusters. The procedure perturbs the residuals of the restricted model (estimated under H_0) at the cluster level using Rademacher weights drawn independently for each cluster, regenerates bootstrap samples under H_0 , and computes a bootstrap p-value as the fraction of bootstrap test statistics exceeding the observed statistic in absolute value. Simulation evidence in Cameron et al. (2008) confirms correct size for $J = 10$ –20 clusters when the regressors of interest are cluster-level, which is precisely the relevant case here since CSAT scores and private tutoring rates vary only across regions, not within them.

As a secondary robustness check, this paper reports CR3 standard errors (Bell and McCaffrey, 2002), which apply a bias-reducing degrees-of-freedom correction to the sandwich estimator, implemented via the `clubSandwich` R package. For the individual-level pooled OLS, the Murphy and Topel (1985) correction addresses the generated regressor problem: since the region-level IGE estimate $\hat{\beta}_j$ is itself estimated from Step 1 and used as the dependent variable in the regional regressions, standard errors that ignore first-stage estimation error understate total uncertainty.

Empirical Strategy

The Parametric Varying-Coefficient Framework

The fixed-coefficient benchmark regresses region-level IGE estimates on regional characteristics:

$$\hat{\beta}_j = c + d_1 \overline{\text{CSAT}}_j + d_2 \text{ShadowRate}_j + d_3 \text{Gini}_j + e_j.$$

This specification assumes that the effect of private education on IGE is the same regardless of a region's development level or exam competition intensity, an assumption that is theoretically implausible and empirically testable. The varying-coefficient generalization allows each coefficient $d_k(\cdot)$ to be a function of the smoothing variable z_j , yielding:

$$\hat{\beta}_j = c(z_j) + d_1(z_j) \overline{\text{CSAT}}_j + d_2(z_j) \text{ShadowRate}_j + d_3(z_j) \text{Gini}_j + e_j.$$

With only $N = 17$ *sido*, full nonparametric kernel estimation is infeasible; Durlauf et al.'s original application used 98 countries. We use the parametric first-order approximation $\beta_k(z_j) \approx \beta_k + \gamma_k z_j$, which, when substituted and expanded, yields a standard OLS regression with interaction terms. The all-varying specification requires estimating ten parameters from seventeen observations, leaving seven residual degrees of freedom — severely limited. The **parsimonious specification**, which allows only the private education coefficient to vary, requires seven parameters and leaves ten degrees of freedom:

$$\ln\left(\frac{y_{j,T}}{y_{j,0}}\right) = \alpha + \beta_1 \ln y_{j,0} + \beta_2 \ln s_j^k + \beta_3 \ln(n_j + g + \delta) + (\beta_4 + \gamma_4 z_j) \ln h_j^{\text{priv}} + \beta_5 \ln h_j^{\text{pub}} + \varepsilon_j.$$

This parsimonious specification is the recommended main result; the all-varying version is reported in the appendix. The null hypothesis $H_0: \gamma_4 = 0$ is the test for parameter heterogeneity: rejection means the private education coefficient in the growth equation varies systematically with exam competition, and by implication the mechanism through which parental income is transmitted to child outcomes differs across regions.

Applied directly to the IGE regression, the same interaction structure becomes:

$$\hat{\beta}_j = c + d_1^0 \overline{\text{CSAT}}_j + d_1^1 (z_j \cdot \overline{\text{CSAT}}_j) + d_2^0 \text{ShadowRate}_j + d_2^1 (z_j \cdot \text{ShadowRate}_j) + d_4 \ln y_{j,0} + d_5 \text{Gini}_j + e_j,$$

where $d_2^1 > 0$ means the private education effect on regional IGE rises with the development level of the region (the amplification hypothesis), and $d_1^1 > 0$ means CSAT competition becomes more distortionary in regions that are already more competitive.

The single most important figure in Step 2 is the $\hat{\beta}_4(z_j)$ curve, plotting $\hat{\beta}_4(z_j) = \hat{\beta}_4 + \hat{\gamma}_4 \cdot z_j$ against z_j with 95 percent bootstrap confidence bands and the seventeen region data points sized proportional to N_j . A horizontal dashed line at zero identifies any threshold development level below which private education has no effect on regional IGE. If the slope is positive, private education is more distortionary in already-competitive regions (amplification); if negative, private education is the binding constraint in less competitive regions (convergence); if the line crosses zero, there is a threshold structure.

Key Variables

The CSAT variables — regional average score $\overline{\text{CSAT}}_j$, regional score variance $\sigma_{\text{CSAT},j}^2$, and top-decile share TopDecile_j — are drawn from KEDI's annual education statistics () and lagged

to the child’s secondary schooling years (birth year + 13 to + 18). An important limitation is that individual CSAT scores are not publicly available; only regional aggregates from KEDI are accessible. This creates an ecological fallacy risk, which is mitigated by including KLIPS individual education level (highest degree: SKY / other four-year / two-year / high school or below) as a control in the individual-level regression. If the regional CSAT interaction remains positive and significant after controlling for individual education level, the finding constitutes evidence of a place effect that operates beyond individual credentials.

Private education variables are drawn from KOSIS’s annual private education expenditure survey () and include participation rate (ShadowRate_{*j*}, the percentage of students attending private tutoring), average years of enrollment per student (ShadowYears_{*j*}), and monthly expenditure per student in Korean won (ShadowSpend_{*j*}). Public education expenditure per pupil (h_j^{pub}) is drawn separately from KOSIS’s local education finance data () and is kept strictly separate from the private education variables throughout all specifications, for the sign-cancellation reasons described in the literature review. Regional GRDP per capita and physical capital investment rates come from KOSIS’s regional accounts (); Gini coefficients from KOSIS’s income distribution indicators (). All regional variables are lagged to the child’s relevant school years using the same birth-year + 13 to + 18 window applied to CSAT.

Steps 1 Through 7

Step 1 (completed) estimates region-level IGE:

$$\ln y_{ij}^c = \alpha_j + \beta_j \ln y_{ij}^p + \mathbf{X}_{ij}'\gamma + \varepsilon_{ij},$$

with standard errors clustered by individual ID. The pre-specified inclusion rule follows Chetty et al. (2014): regions with $N_j < 200$ are excluded from Steps 2–7. This drops Sejong ($N = 32$) and Jeju ($N = 47$) and leaves approximately twelve to fifteen *sido*. The region-level estimates $\hat{\beta}_j$ and their standard errors $\widehat{\text{se}}(\hat{\beta}_j)$ are the inputs to all subsequent steps.

Step 2 estimates the growth-rate IGE regression in DKM estimation form:

$$g_j = a + b_0 \ln y_{j,0} + b_1 \ln s_j^k + b_2 \ln h_j^{pub} + b_3 \ln h_j^{priv} + b_4 \ln(n_j + g + \delta) + u_j,$$

with h^{pub} and h^{priv} entered separately. The regional controls ($\ln s_j^k$, $\ln(n_j + g + \delta)$) are borrowed from the DKM specification as conditioning variables, not to structurally test the Solow model. The role of this step is not to confirm Solow predictions at conventional significance levels — with $K = 5$ regressors and $N = 17$ observations, residual degrees of freedom equal 11, and power is too low to detect meaningful effects reliably — but to verify that the signs of conditional convergence ($b_0 < 0$) and the public versus private education coefficients are consistent with theory, and to produce the diagnostic residual plot. When the residuals $\hat{u}_j$ are plotted against $\ln y_{j,0}$ with a lowess smoother, a systematic positive pattern among high-income regions and negative pattern among low-income regions constitutes the most persuasive single piece of visual evidence for parameter

heterogeneity and motivates the varying-coefficient analysis that follows.

Step 3 is the fixed-coefficient FGLS benchmark, regressing $\hat{\beta}_j$ on $\overline{\text{CSAT}}_j$, ShadowRate_j , ShadowYears_j , $\ln y_{j,0}$, and Gini_j , with FGLS weights $w_j = 1/\widehat{\text{se}}(\hat{\beta}_j)^2$. The expected signs are positive for both CSAT and shadow education variables (tighter exam competition and wider tutoring participation should each raise regional IGE) and positive for the Gini coefficient (following the Galor–Zeira credit constraint mechanism). OLS and FGLS are reported side by side; qualitative similarity between the two suggests heteroskedasticity affected standard errors rather than point estimates, which is the expected outcome with only seventeen observations.

Step 4 estimates the parsimonious varying-coefficient model described above, with $z_j = \overline{\text{CSAT}}_j$ as the main specification and $z_j = \ln y_{j,0}$ as the robustness check. The coefficient $\hat{\gamma}_4$ on the interaction term ($z_j \cdot \ln h_j^{\text{priv}}$) is the primary quantity of interest. A positive estimate supports the amplification hypothesis; a negative estimate supports the convergence hypothesis. If both the main and robustness specifications yield consistent signs on $\hat{\gamma}_4$, the finding is robust to the polynomial equivalence concern.

Step 5 addresses inference. With $J = 12\text{--}17$ clusters, the wild cluster bootstrap (Cameron et al., 2008) is the primary inference method. The procedure estimates the restricted model under $H_0: \gamma_4 = 0$, obtains restricted residuals $\tilde{e}_j$, and for each of $B = 5,000$ bootstrap replications draws Rademacher signs $v_j^{(b)} \in \{-1, +1\}$ independently for each region j , constructs perturbed outcomes $\tilde{Y}_j^{(b)} = X_j' \tilde{\beta} + v_j^{(b)} \cdot \tilde{e}_j$, and re-estimates the full model to obtain bootstrap test statistics $t^{(b)}$. The bootstrap p-value is the share of $|t^{(b)}|$ exceeding $|t_{\text{obs}}|$. The Murphy–Topel (1985) analytical correction for generated regressor bias is reported alongside the bootstrap, and CR3 standard errors (Bell and McCaffrey, 2002) are reported as an additional robustness check.

Step 6 leverages the full individual-level sample ($N = 5,388$) via pooled OLS:

$$\ln y_{ij}^e = \alpha + \beta_1 \ln \bar{y}_i^p + \beta_2 \text{ShadowRate}_j + \beta_3 (\text{ShadowRate}_j \times \ln \bar{y}_i^p) + \beta_4 \overline{\text{CSAT}}_j + \beta_5 (\overline{\text{CSAT}}_j \times \ln \bar{y}_i^p) + \mathbf{X}_i' \delta + \mu_j + \varepsilon_{ij}.$$

Two models are estimated. Model A excludes region fixed effects and estimates both the level effects β_2, β_4 and the interaction effects β_3, β_5 , at the cost of potential bias from time-invariant regional confounders. Model B includes region fixed effects μ_j , which absorb ShadowRate_j and $\overline{\text{CSAT}}_j$ entirely since they are time-invariant at the regional level, leaving only the interaction coefficients β_3 and β_5 identified. Model B is the preferred specification: the region fixed effects absorb all time-invariant regional confounders, and $\hat{\beta}_3$ and $\hat{\beta}_5$ identify whether higher parental income amplifies the advantage of living in a high-tutoring or high-CSAT region. Wild cluster bootstrap p-values and CR3 standard errors are reported for both key coefficients. Individual education level (SKY / other four-year / two-year / high school or below) is included as a control; if the CSAT interaction $\hat{\beta}_5$ survives this control, the estimate reflects a place effect beyond individual credentials.

Step 7 extends the quantile regression from Step 1 to the regional dimension. With 17 *sido* and five quantile points, the average cell contains only $N \approx 63$ observations, and cells for small regions at extreme quantiles may have fewer than ten. To address this, the 17 *sido* are aggregated into four pre-specified regional groups: Capital (Seoul, Gyeonggi, Incheon; $N \approx 5,453$), Metropolitan

(Busan, Daegu, Gwangju, Daejeon, Ulsan; $N \approx 2,126$), Province-large (Gyeongnam, Gyeongbuk, Chungnam, Jeonnam; $N \approx 1,541$), and Province-small (remaining *sido*; $N \approx 706$). Each group provides sufficient observations for stable quantile estimates across $\tau = 0.10, 0.25, 0.50, 0.75, 0.90$ with empirical bootstrap standard errors. The key figure plots $\hat{\beta}(\tau)$ by regional group across quantiles. The two scenarios of primary interest are whether bottom-quantile Capital IGE substantially exceeds Province IGE (a “glass ceiling for the poor” in which private tutoring is most necessary and most expensive precisely where the income distribution is most unequal) and whether top-quantile Capital IGE converges with Province IGE (a “glass floor” in which wealthy families protect their children’s positions equally across regions regardless of tutoring market intensity). These scenarios carry fundamentally different policy implications.

Hypotheses

The following seven hypotheses correspond directly to the steps above. H1 and H2 are tested by the fixed-coefficient FGLS benchmark (Step 3): the private education participation rate should raise regional IGE ($d_2 > 0$), and higher regional CSAT competition should raise regional IGE ($d_1 > 0$). H3 and H4 are tested by the varying-coefficient model (Step 4): the private education effect on IGE should rise with the smoothing variable ($d_2^1 > 0$, amplification), and the CSAT effect on IGE should similarly rise with z_j ($d_1^1 > 0$). H5 is tested by the residual plot from Step 2: private education should weaken intergenerational mean reversion (conditional convergence in the growth-rate form), visible as a positive systematic pattern in residuals against initial income for high-CSAT regions. H6 is tested within Step 6 by comparing the shadow education interaction coefficient $\hat{\beta}_3$ between low-quintile and high-quintile subsamples of the income distribution; the prediction is that the shadow education effect is stronger at the bottom. H7 is tested by Step 7: the bottom-quantile IGE gap between the Capital and provincial groups should exceed the top-quantile IGE gap, reflecting the asymmetric impact of private tutoring market intensity on immobility at the lower tail.

Limitations

This analysis faces several important limitations that must be acknowledged. First, the strategy is observational and does not use instrumental variables or natural experiments. Unobserved time-invariant regional characteristics — institutional quality, commuting network density, cultural attitudes toward credentials — may confound the estimates even in the fixed-effects specification. The contribution is a theoretically motivated, mechanistically interpretable conditional correlation, not a causal claim.

Second, with only $N = 17$ regional units, the full nonparametric kernel estimation of DKM is infeasible, and the parametric first-order approximation cannot detect nonlinear variation (U-shaped or threshold patterns) in the IGE coefficients. The polynomial equivalence concern when $z_j = \ln y_{j,0}$ is addressed by the preferred specification using $z_j = \overline{\text{CSAT}}_j$, but the parametric

approximation remains a restriction on the functional form of parameter heterogeneity.

Third, individual CSAT scores are not publicly available; only regional aggregates from KEDI are used. Measurement error in $\overline{\text{CSAT}}_j$ will attenuate $\hat{d}_1$ and $\hat{\beta}_5$ toward zero, making estimates of H2 and H4 conservative. Including individual education level as a control mitigates the ecological fallacy concern but cannot eliminate it entirely.

Fourth, $\hat{\beta}_j$ carries Step 1 estimation error into Steps 3 and 4 as a generated regressor. The Murphy–Topel correction and bootstrap address this, but the correction is approximate rather than exact, and with only seventeen region-level observations any mechanical correction must be interpreted cautiously.

Fifth, the DKM framework is adopted here solely as an estimation strategy for a varying-coefficient IGE regression. The structural content of the Solow model — capital accumulation, savings rates, population growth as determinants of steady-state income — is not maintained. The variables $\ln s_j^k$ and $\ln(n_j + g + \delta)$ enter as empirical controls rather than as tests of Solow theory, and the convergence coefficient b_0 is interpreted as mean reversion in the IGE sense, not as Solow conditional convergence.

Sixth, and most fundamentally, Seoul’s private education boom may have general equilibrium effects on the competitive threshold for university admission that depress the relative CSAT performance of provincial students even without any change in their own tutoring investment — a SUTVA violation that no single-region or cross-region regression model can address.

The appropriate framing for the empirical strategy as a whole is as follows. We adopt the varying-coefficient regression framework of Durlauf, Kourtellos, and Minkin (2001) as an estimation strategy, not as a structural model of capital accumulation. The connection is algebraic: subtracting initial log income from both sides of the standard log-log IGE equation $\ln y_{i,t} = b(Z_j) \ln y_{i,t-1} + \varepsilon_{i,t}$ yields the growth-rate form $g_{i,t} = c(Z_j) \ln y_{i,t-1} + \varepsilon_{i,t}$, where $c(Z_j) = b(Z_j) - 1$. This is mathematically identical to the cross-regional regression estimated by DKM, with the coefficient on initial income playing the role of the varying convergence parameter. The smoothing variable Z_j in our application is regional CSAT performance — a theoretically motivated determinant of the private education mechanism — rather than initial per-capita income as in DKM. We address the small-sample inference problem via wild cluster bootstrap (Cameron et al., 2008) and the generated-regressor problem via Murphy–Topel (1985) correction. While this strategy does not constitute causal identification, it provides a mechanistically motivated and methodologically disciplined account of the spatial heterogeneity in Korean intergenerational mobility.

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